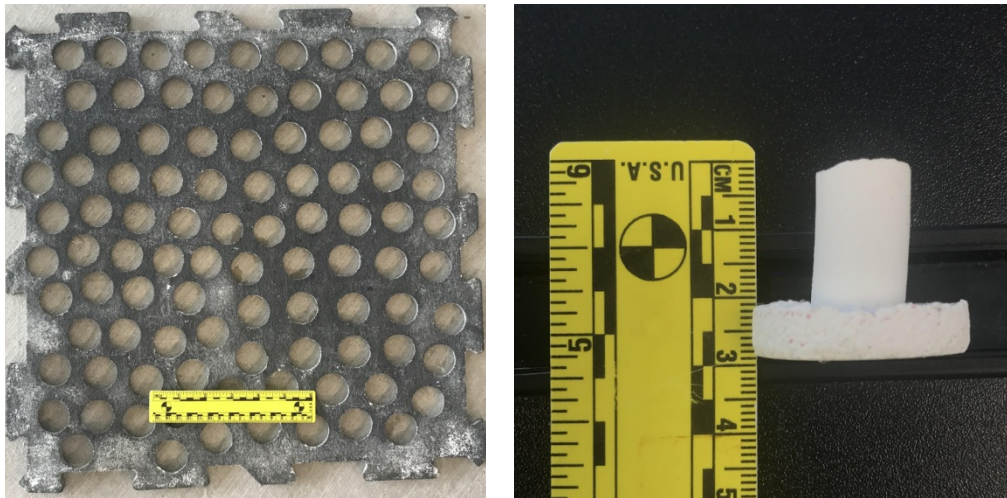
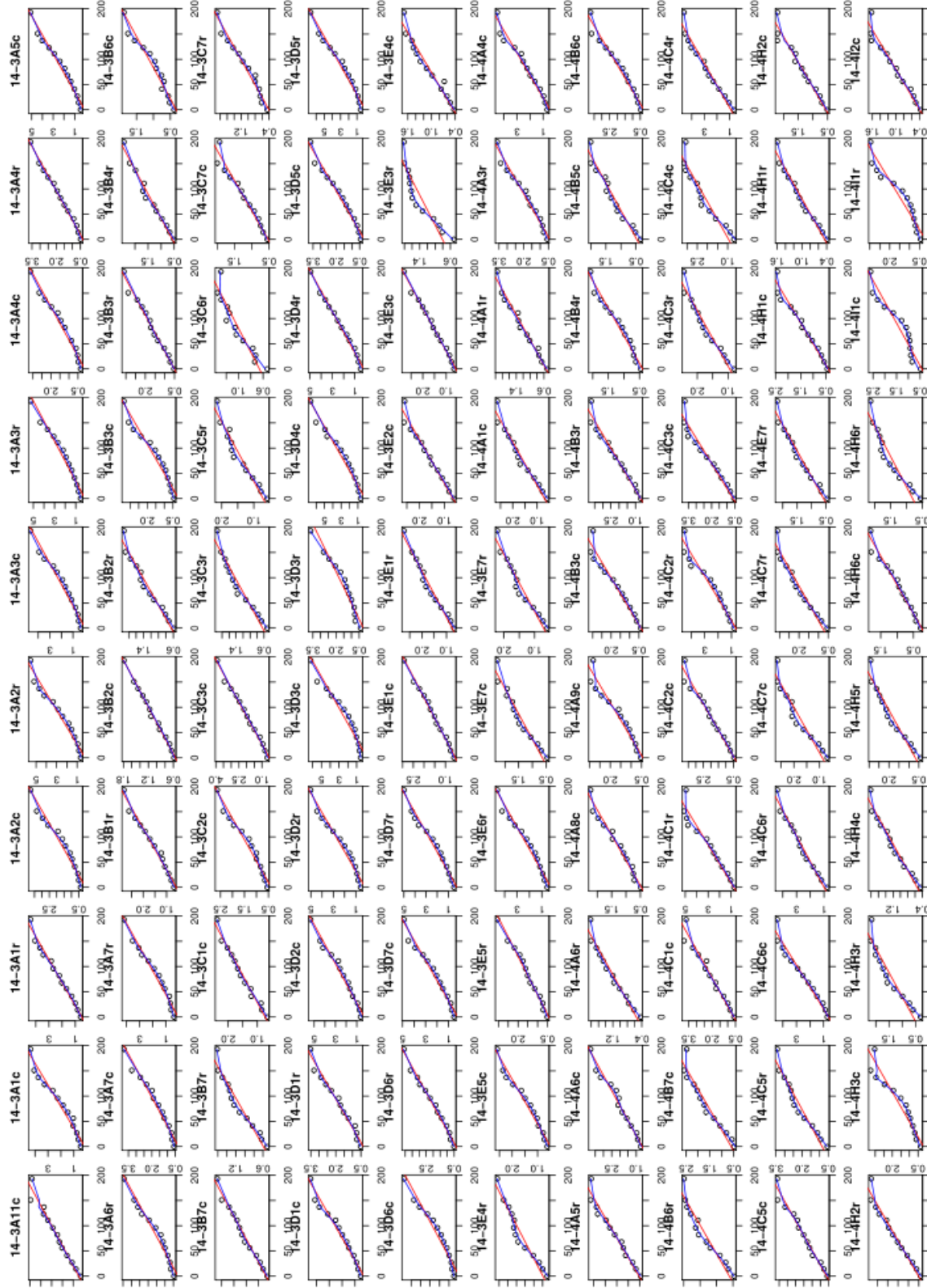


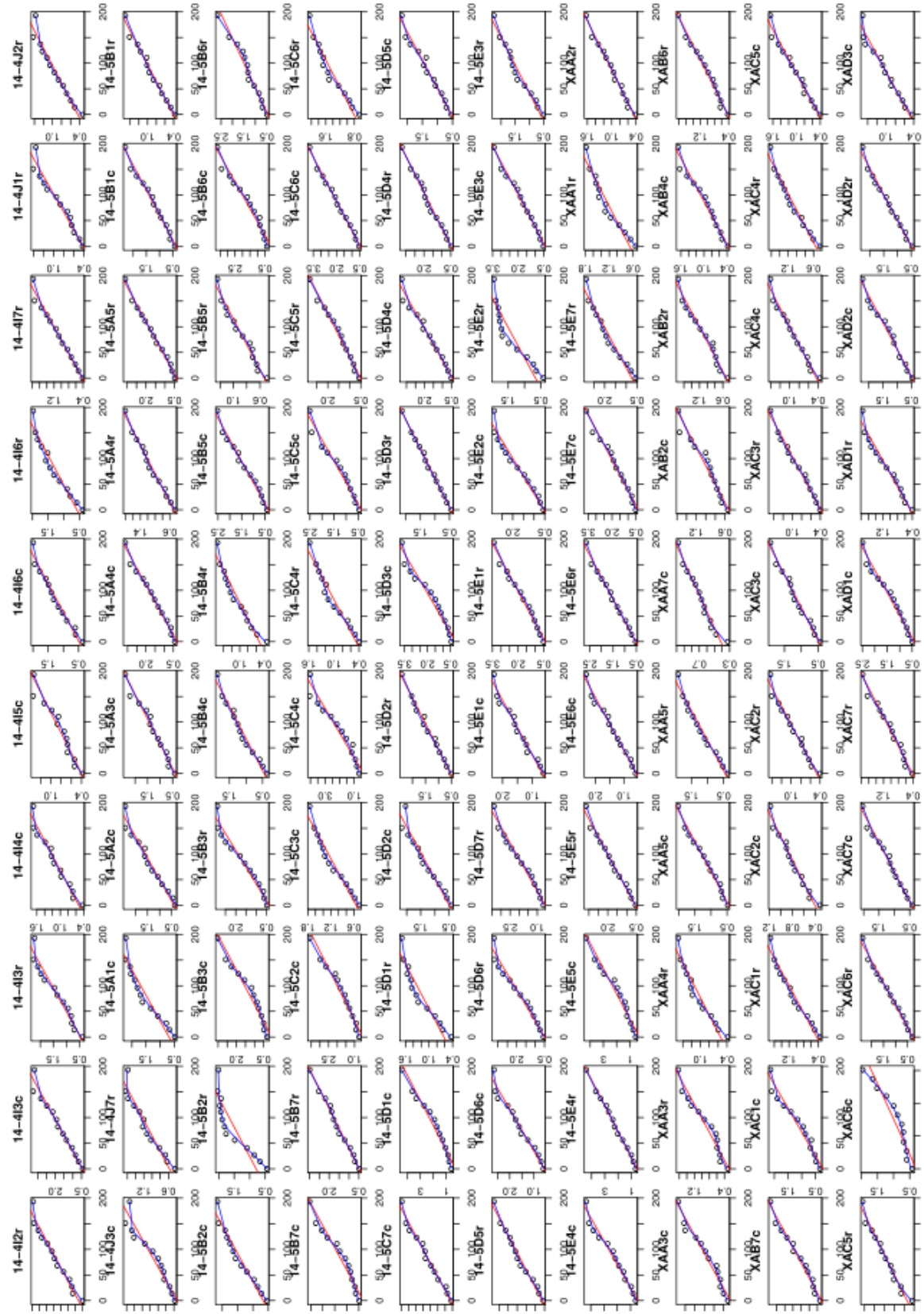
Cement plug construction

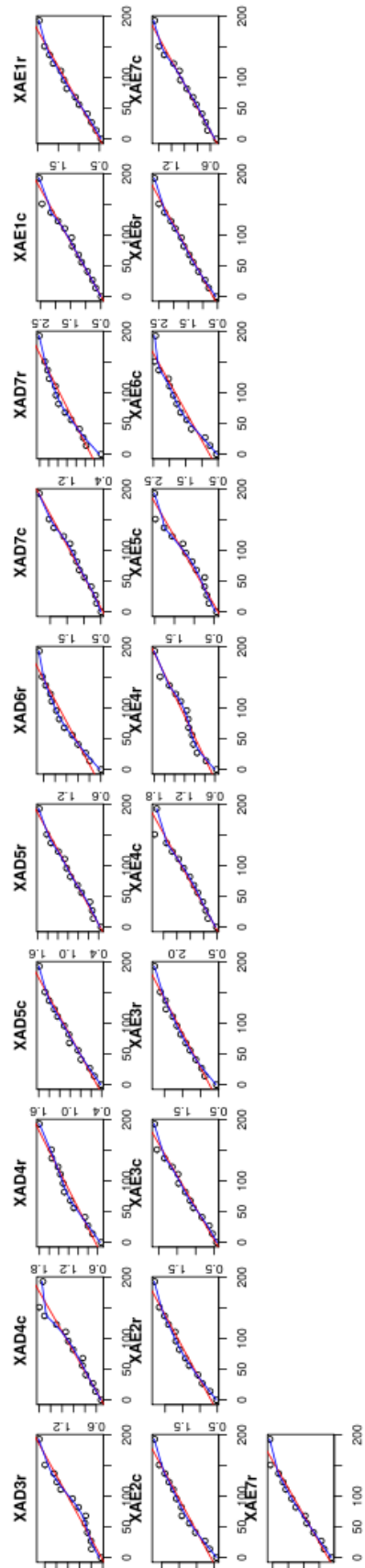
Cement plugs (stem and disk) were constructed from a 3:12:7 ratio mix of commercially available playground sand, white portland cement, and freshwater. Stems were made by pouring cement into 13 mm (inside diameter) flexible aquarium tubing sliced open vertically and held together with tape. Stems were allowed to set overnight and once set, the cement was removed from the tubing by opening the tape. A band saw was used to cut the cement tube into ~2 cm long posts. A 0.5 cm deep rubber mat was drilled with a 3 cm diameter spade bit to create the disc mold (Figure S3). Cement was poured into each hole and flattened using a squeegee to create a smooth disc surface. Pre-cut stems were then gently pushed into the wet cement to complete the ~2.5 cm tall plug (Figure S1). Plugs were allowed to harden overnight and then placed in a 5 gallon bucket of fresh water to soak for 6 weeks. Fresh water was refreshed regularly.



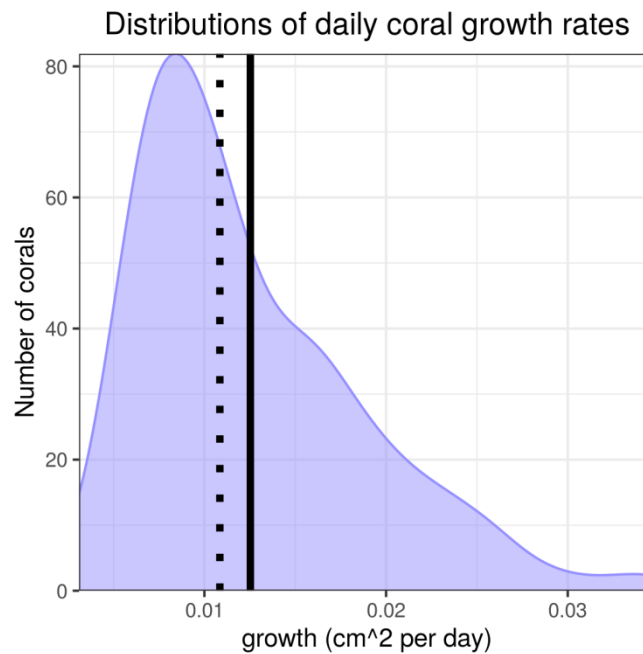
Supplementary Figure S1. (left) Cement plug mold created by drilling 3 cm diameter holes in 0.5 cm deep rubber mats; (right) Cement plugs (stem and disk) made for growing coral.







Supplementary Figure S2. Individual coral growth curves. Least squares regression (red) and LOESS smoother (span =0.5, blue) used to illustrate growth patterns.



Supplementary Figure S3. Density plot of untransformed coral growth rates used in this experiment. Dotted and solid lines indicate median and mean growth rates, respectively. Data were log10 transformed prior to analysis to approximate normality of residuals.

Supplementary Table S1. Number of coral genotypes, parent colonies, and microfragments used in the experiment, split by substrate type.

Genotype ID	14-3	14-4	14-5	XA	Total
# Ramets fragmented	5	7	5	5	22
# Microfragments attached to cement plugs	28	28	30	26	112
# Microfragments attached to ceramic	28	28	26	27	109

Supplementary Table S2. Models evaluated via nlme in the experiment. The colon denotes an interaction between the two main effects, while the | operator indicates the ramet random effect was allowed to vary based on levels of the other factor included. The dredge function also

included an intercept only model which was not evaluated here. Model selection methods followed Zuur et al. 2009 with random effects selection first, followed by fixed effects evaluation after optimal random effects structure was determined. The optimal random effects model and fixed effects model are bolded.

Analysis stage	Model Fixed Effects	Random Effects
<i>Random Effects Selection</i>	<i>Growth ~ Geno + Sub + Geno:Sub</i>	<i>None</i>
<i>Random Effects Selection</i>	<i>Growth ~ Geno + Sub + Geno:Sub</i>	<i>Ramet</i>
<i>Random Effects Selection</i>	<i>Growth ~ Geno + Sub + Geno:Sub</i>	<i>Substrate Ramet</i>
<i>Random Effects Selection</i>	<i>Growth ~ Geno + Sub + Geno:Sub</i>	<i>Genotype Ramet</i>
<i>Fixed Effects Selection</i>	<i>Growth ~ Geno</i>	<i>Ramet</i>
<i>Fixed Effects Selection</i>	<i>Growth ~ Sub</i>	<i>Ramet</i>
<i>Fixed Effects Selection</i>	<i>Growth ~ Geno + Sub</i>	<i>Ramet</i>
<i>Fixed Effects Selection</i>	<i>Growth ~ Geno + Sub + Geno:Sub</i>	<i>Ramet</i>